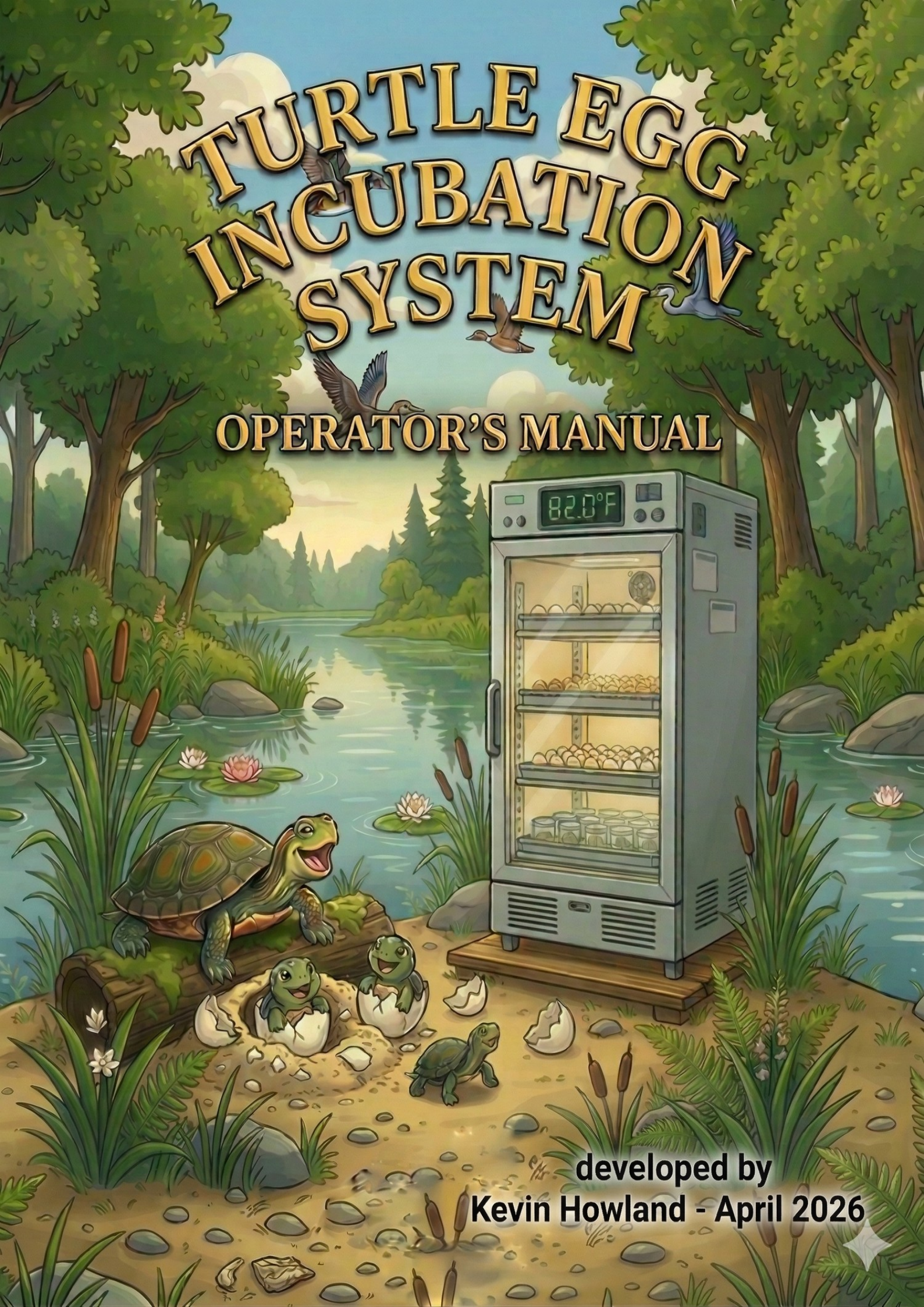
TURTLE EGG INCUBATION SYSTEM

OPERATOR'S MANUAL



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WINC Incubator System

Operator's Guide and Clinical Protocol

Release v10.5.1 | Institutional Edition

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1. SIGN-IN: STARTING YOUR SHIFT

Recording clinical data begins with identifying yourself to the database. This ensures that every entry is linked to a specific observer for accountability.

- Select Your Name: Pick your name from the observer registry.
- Sign-In: Click [START] to begin your session.
- The 4-Hour Rule: The system keeps you signed in for 4 hours. After 4 hours of inactivity, you must sign in again to start a new shift. This prevents unauthorized entries under your name if a tablet is left unattended.

1.3 Documentation Conventions

To help you find information fast, we use a specific "Visual Logic" system:

INTRODUCTION TO THE SECURE DATA NETWORK

The WINC System is a secure, high-integrity database designed to protect turtle research data from loss or corruption.

1.4 The Three Pillars of the System

To use WINC correctly, you must understand how the pieces fit together:

This is where the actual records live. When you click [SAVE], your data travels to a high-security container. Every piece of data has a "Timestamp" and a "Signature" (your name).

This is the simple screen you use on your tablet or phone. It has been designed for Field Simplicity, filtering out the noise so you can focus only on the eggs in front of you.

A background system that logs all changes for accuracy. If you correct an entry, the audit system keeps a copy of what you changed and why you changed it.

1.5 System Hierarchy: From Machine to Egg

WINC follows a "Family Tree" logic: 1. The Incubator: The large physical machine in the lab. 2. The Bin: The plastic box inside the machine. 3. The Mother: The turtle that provided the eggs. 4. The Egg: The individual life we are protecting.

2. INTAKE: ADDING NEW MOTHERS AND EGGS

When a mother turtle is found and eggs are collected, we must add the new records to the database. This creates a permanent digital history for the mother, her plastic bin, and every individual egg.

2.1 Standard Clinical Intake Workflow

Follow these steps to register a new discovery:

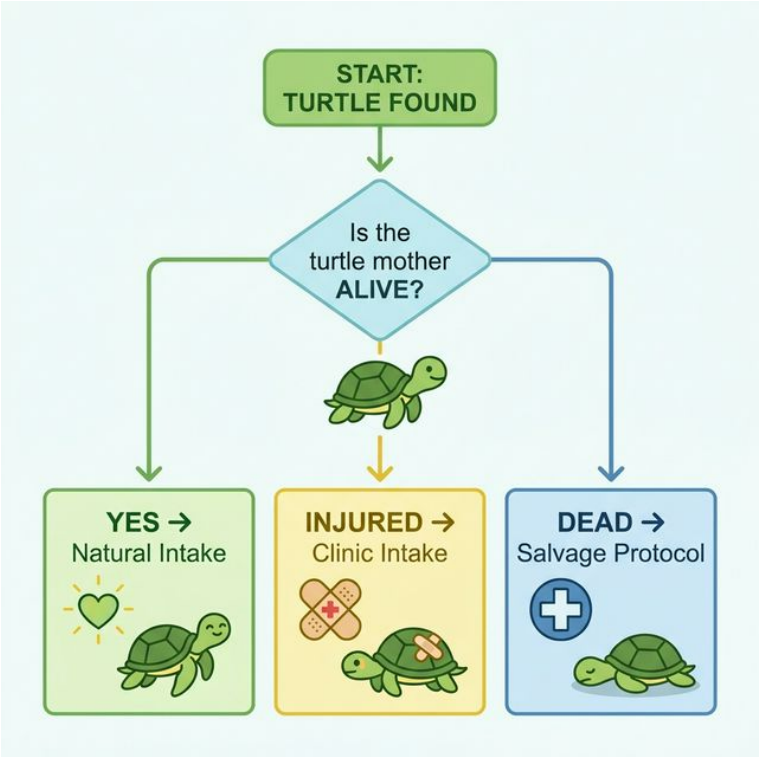
- Species Identification: Select the correct turtle type (e.g., Blandings or Wood Turtle).
- Assign WINC Case #: Type the year and the sequence number (e.g., 2026-003).
- Discovery Details: Type the Finder's Name and where they found the turtle.
- Automatic Record Creation: Click [SAVE]. The database will automatically create 1 Mother record, 1 Bin record, and all individual egg records at once.

2.2 Multiple Bins for one Mother

If a mother turtle lays more eggs than will fit in one bin: 1. Complete the first intake for Bin 1. 2. Use the ADD function in the sidebar to link a second bin to the same Mother ID.

3.1 Choosing the Intake Path

Not every turtle arrives in the same way. Use the flowchart below to decide which path to take before you touch the screen.



3.2 Standard Clinical Intake Workflow

Follow these steps to register a healthy discovery:

- Species Identification: Click the Species box. Choose the correct turtle type (e.g., Blandings or Wood Turtle). The system uses this to set the "Optimal Incubation Temp" alerts later.
- Assign WINC Case #: Type the year and the sequence number (e.g., 2026-003). This is the "Master ID" for the mother.
- Discovery Details: Type the Finder's Name and where they found the turtle.
- Maternal Metrics: Type the Mother's Weight (g) and Carapace Length (mm) if required by the Lead Biologist.
- Establishing Case: Double-check your numbers. Then, click the big [SAVE] button.

NOTE: When you click [SAVE], the "Audit Robot" creates 1 Mother record, 1 Bin record, and up to 30 individual Egg records all at once. This is called an Atomic Transaction.

3.3 The DOA Salvage Protocol

If a turtle is found deceased (Dead on Arrival) but the eggs are still viable, follow this path: 1. Set the Condition to Dead (Salvage). 2. Complete the same steps as above, but add a NOTE in the Intake Notes field about the location of the recovery. 3. The system will flag this case as a Salvage Recovery in the final reports.

3.4 Multiple Bins for one Mother

Sometimes a mother turtle lays so many eggs that they won't fit in one plastic bin. 1. Complete the first intake for Bin 1. 2. Go to the Supplemental Intake tools in the sidebar. 3. Choose the same Mother ID and click [ADD] to create Bin 2.

IMPORTANT: MOTHER ID VERIFICATION: Before clicking [SAVE], look at the physical tag on the mother turtle. Ensure it matches the Mother ID you selected in the app. Choosing the wrong ID will link your 30 eggs to the wrong biological parent in the permanent cloud records.

3. DAILY CHECKS: RECORDING EGG HEALTH

This screen is used for the daily monitoring of bin environment and individual egg growth. Every bin must be weighed before individual eggs are handled.

3.1 The Weighing Protocol (Clinical Unlock)

- Mass Verification: Take the physical bin to the scale. Type the weight in the Current Total Mass (g) box.
- Compare to Target: Look at the Target Weight listed on the screen.
- Remediation Action: Weight is Low: Add water (ml) to match the Target. Enter the amount in Water Added. Weight is High (>5g Over): Do not add water. Open the bin lid for 2 hours for passive evaporation. Weight is Very High (>15g Over): Immediately notify the Lead Biologist for a substrate swap.
- Weight is Low: Add water (ml) to match the Target. Enter the amount in Water Added.
- Weight is High (>5g Over): Do not add water. Open the bin lid for 2 hours for passive evaporation.
- Weight is Very High (>15g Over): Immediately notify the Lead Biologist for a substrate swap.
- Clinical Unlock: Click [SAVE] to unlock the individual egg grid.

3.2 Biological Mandate: NEVER ROTATE

IMPORTANT: When you pick up an egg to check it, NEVER TURN IT OVER. Flip-risk: If you flip the egg, the embryo may drown in its own yolk.

3.3 Recording Health (The 0-3 Scale)

- Select Eggs: Select individual eggs or use the [START] button to select all pending.
- Assign Stage: Pick the current growth stage (e.g., S3M).
- Set Health Scores: Rate health markers from 0 (Normal) to 3 (Critical).
- Save: Click [SAVE] at the bottom to sign and record your work.

3.4 Species-Specific Alert Thresholds

The database automatically flags bins that fall outside these safe ranges. Use this table to double-check system alerts:

4. LIFECYCLE: EGG DEVELOPMENT STAGES (S0 TO S6)

Turtle eggs move through seven development stages from "Just Laid" (S0) to "Hatched" (S6).

4.1 Understanding Stage Codes

- The Number (0-6): The lifecycle stage.
- The Letter (S, M, J): The progress level (Small, Medium, or Major).

4.2 Recording Hatching Vitality (Stage S6)

When an egg reaches S6, you must record a Vitality Score (0-5) to determine the next action: - Score 5 (Strong): Baby is fully active. Action: Ready for transfer to release bin. - Score 3 (Guarded): Slow movement or large yolk sac. Action: Move to Medical Observation. - Score 0 (Failed): Baby died during hatching. Action: Notify Senior Biologist immediately.

5. ADMIN: FIXING ERRORS AND SETTINGS

5.1 Correction Mode

To keep records accurate, the system never fully deletes an entry. It marks it as "Voided" and saves a copy of the original for the background audit system.

- Toggle Correction Mode to ON at the top of the screen.
- Find the record and click [VOID].
- Type a reason for the change (Mandatory).

5.2 Deleted Records Folder

If a bin was accidentally deleted while still containing active eggs, it is moved to the Deleted Records Folder. Contact an Administrator to restore these records.

6. REPORTS: READING THE CLINICAL DATA (ANALYTICS)

The WINC Vault doesn't just store data; it translates it into biological stories. These reports help us understand which clutches are healthy and which ones need more help.

6.1 The Mortality Heatmap

This is the most important report for the Science team. It shows a grid of "Danger Zones." * Green Squares: Normal development for that time of year. * Red Squares: A "Critical Window" where many eggs are failing. If you see a cluster of Red, check the incubator temperature immediately!

[Figure 6.1: Report Placeholder - Mortality Heatmap View]

6.2 WormD Exports (Sharing for Science)

WINC is part of a global network. We share our data with other turtle scientists using a format called WormD. 1. Go to the Download Data screen. 2. Select WormD Export. 3. The system will create a "Clean" file that removes sensitive location data but keeps the biological health scores.

7. CRISIS: WHEN THE INTERNET FAILS (CONTINUITY)

Field tablets rely on Wi-Fi or Cellular data. If the connection drops in the middle of a check, follow this Resilience Protocol.

7.1 The Resilience Protocol

- DO NOT REFRESH: If you refresh the browser while the internet is "Red," you will lose the data currently in your tablet's local memory.
- Switch to Paper: Immediately pull out your Physical Field Sheets (v2026). Record all weights and health scores by hand.
- The Double-Witness Rule: Once the internet is restored, you must type the paper logs into the software. A second person (the "Witness") must sit next to you and check your typing against the paper to ensure there are no Subject ID typos.

8. GLOSSARY

- Automatic Record Creation: The background system that creates egg records immediately after a Mother Intake is saved.
- Chalking: The white calcium patches on the shell indicating embryo development.
- Correction Mode: A secure state that allows observers to void incorrect entries.
- Local Data Storage: The fact that our data lives in our local database, not a third-party commercial cloud.
- Egg: An individual developing turtle subject.
- Sign-In: The process of identifying yourself to the database to start a shift.
- Voiding: Marking an entry as incorrect without deleting it from the audit log.
- WormD: The standard scientific format for exporting turtle health data.

WINC Clinical Documentation 2026 Platinum Rated. Clinical Excellence. Failure-Proof.